

# Countdown 31 — Comprehensive Backend Engineering Specification

Definitive architectural contract, database schema migrations, real-time gateway protocols, REST APIs, sponsor release controls, template registries, telemetry ingestion pipelines, and multi-tenant security requirements for full production deployment.

**Target Repository:** emustwe/countdown-31 · **Stack:** NestJS 11 · Prisma 6 · PostgreSQL 16 · Redis 7 · Socket.IO · TypeScript  
**Engineering Lead / Author:** Muhammad Sameer Ali <SameerAli126@users.noreply.github.com> · **Date:** August 27, 2026  
**Scope:** Complete Real-World Implementation Guide for Admin Panel, Tournament Platform, Campaign Studio, and Arena Integration.

## EXECUTIVE MANDATE & IMPLEMENTATION OBJECTIVE:

This document serves as the single source of truth for the backend engineering team. It details every database relation, business rule, REST endpoint, WebSocket gateway event, background cron job, telemetry batch ingestion worker, and file storage requirement needed to operate the verified Next.js frontend client in a real-world, high-concurrency production environment without relying on mock data or local fallbacks.

## 1. System Architecture, Runtime Domains & Security Personas

The platform operates as a distributed system with three distinct operational runtime layers and RBAC boundaries:

| Runtime Persona / Surface                                                                                                           | Responsibilities & Real-World Capabilities                                                                                                                                                                                                                                                                                                                          | Auth & Security Guards                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Super Admin Console</b><br><code>/admin</code><br><code>/admin/tournament</code><br><code>/admin/tournament/[id]/campaign</code> | Full CRUD management of solo knockouts & influencer group tournaments; gate code generation (sponsorCode, joinCode, captainCode, memberCode); campaign templates registry CRUD (Nike 31, Adidas 31, MooMorrow, etc.); multi-lane approval pipeline (BRAND & SAFETY); version publishing, 1-click rollback, and emergency kill switches (isPaused, isCausePaused).   | JWT + Role.ADMIN<br>AdminGuard with tokenVersion revocation check; TOTP 2FA enforcement.                           |
| <b>Sponsor Self-Service Portal</b><br><code>/sponsor/login</code><br><code>/admin/tournament/[id]/campaign</code>                   | Self-service customization of co-branded visual assets; upload logos (PNG/SVG/MP4), desktop arena backgrounds, and mobile backgrounds; configure charity cause details (target amount, beneficiary, currency, CTA); submit drafts for admin review; inspect live privacy-safe telemetry and download proof-of-performance CSV reports.                              | SponsorAuthGuard<br>Isolated sponsor accounts; token scoped exclusively to owned tournaments via sponsorId.        |
| <b>Player Client Experience</b><br><code>/home</code> , <code>/events</code><br><code>/events/[id]</code> , <code>/wallet</code>    | Browse public tournaments; enter private tournaments via referral codes; participate in GMT start-time slot voting; join real-time match lobbies; experience dynamic [Sponsor] 31 branding (Nike 31, Adidas 31, MooMorrow 31) across header, arena HUD, and mobile modals; track charity cause milestones; play cards 1-3, use skills, and compete for prize pools. | AuthGuard or Guest Session.<br>Rate-limited turn submission;<br>Solana signature verification for crypto deposits. |

## 2. Complete PostgreSQL Database Schema & Entity Relational Model (Prisma ORM)

The backend must maintain the following relational schema in PostgreSQL via Prisma 6. All models include required indexes, unique constraints, cascade deletion behaviors, and audit timestamps:

| Prisma Model             | Field Definitions & Column Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Relations & Indexes                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SponsorTournament</b> | id (UUID pk), title (VarChar), description (Text), visibility (PUBLIC PRIVATE), status (PENDING APPROVED LIVE COMPLETED), type (REGULAR INFLUENCER), groupCount (Int def:2), hasInfluencers (Bool def:false), minPlayers/maxPlayers (Int?), minGroupPlayers/maxGroupPlayers (Int?), startDate (Date?), timeOptions (Json def:[]), startAt (DateTime?), endAt (DateTime?), prizePool (VarChar def:""), winnerCount (Int def:1), seekingSponsor (Bool def:false), sponsorCode (VarChar? unique), joinCode (VarChar? unique), sponsorId (UUID?), createdBy (VarChar), createdAt (DateTime), updatedAt (DateTime) | sponsor -> Sponsor?<br>campaign -> TournamentCampaign?<br>teams -> TournamentTeam[]<br>timeVotes -> TournamentTimeVote[]<br>entries -> PromoEntry[]<br>Indexes: [status], [visibility, status], [startAt] |

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                       |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TournamentTeam</b>                                                  | id (UUID pk), tournamentId (UUID fk), name (VarChar), captainName (VarChar), captainCode (VarChar unique), memberCode (VarChar? unique), color (VarChar def:#5aa8ff), createdAt (DateTime)                                                                                                                                                                                                                           | tournament -> SponsorTournament (onDelete: Cascade)<br>entries -> PromoEntry[]<br>Index: [tournamentId]                                                                               |
| <b>TournamentTimeVote</b>                                              | id (UUID pk), tournamentId (UUID fk), userId (UUID fk), slot (VarChar e.g. '18:00'), createdAt (DateTime)                                                                                                                                                                                                                                                                                                            | tournament -> SponsorTournament<br>user -> User<br>Unique: [tournamentId, userId]<br>Index: [tournamentId, slot]                                                                      |
| <b>TournamentCampaign</b>                                              | id (UUID pk), tournamentId (UUID fk unique), isPaused (Bool def:false), isCausePaused (Bool def:false), createdAt (DateTime), updatedAt (DateTime)                                                                                                                                                                                                                                                                   | tournament -> SponsorTournament (onDelete: Cascade)<br>versions -> TournamentCampaignVersion[]<br>assets -> TournamentCampaignAsset[]<br>events -> TournamentCampaignEventAggregate[] |
| <b>TournamentCampaignVersion</b>                                       | id (UUID pk), campaignId (UUID fk), revision (Int), status (DRAFT IN_REVIEW APPROVED PUBLISHED ARCHIVED), manifest (Json), createdBy (VarChar), createdAt (DateTime), publishedAt (DateTime?), approvedBy (VarChar?), approvedAt (DateTime?), activateAt (DateTime?), expireAt (DateTime?)                                                                                                                           | campaign -> TournamentCampaign (onDelete: Cascade)<br>reviews -> TournamentCampaignReview[]<br>Unique: [campaignId, revision]<br>Index: [campaignId, status]                          |
| <b>TournamentCampaignReview</b>                                        | id (UUID pk), versionId (UUID fk), reviewerId (VarChar), lane (BRAND SAFETY), decision (COMMENT APPROVED CHANGES_REQUESTED), comment (Text), createdAt (DateTime)                                                                                                                                                                                                                                                    | version -> TournamentCampaignVersion (onDelete: Cascade)<br>Index: [versionId, createdAt]                                                                                             |
| <b>TournamentCampaignAsset</b>                                         | id (UUID pk), campaignId (UUID fk), kind (LOGO BACKGROUND_DESKTOP BACKGROUND_MOBILE), url (VarChar), storagePath (VarChar), originalName (VarChar), mimeType (VarChar), bytes (Int), createdBy (VarChar), supersedesId (UUID?), archivedAt (DateTime?), createdAt (DateTime)                                                                                                                                         | campaign -> TournamentCampaign (onDelete: Cascade)<br>Index: [campaignId, kind, createdAt]                                                                                            |
| <b>TournamentCampaignTemplate</b><br><i>(Global Template Registry)</i> | id (UUID pk), slug (VarChar unique), name (VarChar), category (VarChar), badge (VarChar), description (Text), manifest (Json), isPreset (Bool def:false), createdBy (VarChar?), createdAt (DateTime), updatedAt (DateTime)                                                                                                                                                                                           | Stores global sponsor presets (Nike 31, Adidas 31, etc.) & custom templates.<br>Index: [isPreset, category]                                                                           |
| <b>TournamentCampaignEventAggregate</b>                                | id (UUID pk), campaignId (UUID fk), tournamentId (VarChar), revision (Int), placement (logoTile arenaBackground featurePanel causeCard lobbyHero resultSignature), deviceClass (desktop tablet mobile), eventType (eligible_load rendered_impression viewable_seconds completed_loop cause_expand cta_click), count (Int def:0), totalSeconds (Float def:0), dateBucket (VarChar 'YYYY-MM-DD'), createdAt, updatedAt | campaign -> TournamentCampaign (onDelete: Cascade)<br>Unique: [campaignId, revision, placement, deviceClass, eventType, dateBucket]<br>Indexes: [tournamentId, dateBucket]            |
| <b>PromoEntry</b>                                                      | id (UUID pk), tournamentId (UUID fk), userId (UUID fk), teamId (UUID? fk), score (Int def:0), rank (Int?), prizeAwarded (VarChar?), createdAt (DateTime)                                                                                                                                                                                                                                                             | tournament -> SponsorTournament<br>user -> User<br>team -> TournamentTeam?<br>Unique: [tournamentId, userId]                                                                          |

### 3. Campaign Manifest JSON Data Structure & Validation Rules

The manifest JSON stored in TournamentCampaignVersion.manifest and TournamentCampaignTemplate.manifest must pass strict schema validation before persistence:

```
{
  "identity": {
    "campaignTitle": "Nike 31 Speed Championship", // Display title across banners & lobby
    "sponsorName": "Nike", // Sponsor entity name (e.g. Nike, Adidas, MooMorrow)
    "disclosureLabel": "Powered by", // "Presented by" | "Powered by" | "Sponsored by"
    "demoDisclaimer": "Official Partner Series 2026" // Subtitle disclaimer text
  },
  "theme": {
    "primaryColor": "#ffffff", // Dominant typography & title text color
    "secondaryColor": "#ccff00", // Accent glow, borders, 3D badges (e.g. Neon Volt)
    "backgroundImage": "/uploads/campaigns/bg.jpg", // Desktop 16:9 arena background image/video
    "mobileBackgroundImage": "/uploads/bg-m.jpg", // Mobile 9:16 portrait viewport background
    "overlayOpacity": 0.75 // Dark vignette / tint opacity (0.0 to 1.0)
  },
  "logoTile": {
    "enabled": true, // Master toggle for header marquee sponsor card
    "logoText": "NIKE", // Fallback text if media fails or is absent
    "animationPreset": "pulse", // "pulse" | "turntable" | "float" | "static"
    "desktopEnabled": true, // Render on desktop header
    "mobileEnabled": true, // Render on mobile compact header
    "mediaUrl": "/uploads/campaigns/logo.png", // Relative storage path or absolute CDN URL
    "mediaType": "image" // "image" (PNG/SVG/WebP) | "video" (MP4/WebM)
  },
  "featurePanel": {
    "enabled": true, // Branded sidebar & lobby card feature box
    "headline": "JUST PLAY. NEVER STOP.", // High-impact promotional headline
    "body": "Precision moves and unstoppable speed. Level up your game and empower young athletes."
  },
  "cause": {
    "enabled": true, // Charity cause tracking toggle
    "label": "Youth Sports Access", // Category badge (e.g. "Youth Sports Access")
    "title": "All-Star Gear & Safe Playgrounds Fund", // Specific cause project title
    "message": "Providing pro equipment and renovated courts for kids in underserved neighborhoods.",
    "beneficiaryName": "Global Youth Athletic Alliance", // Non-profit organization / trust name
    "targetAmount": 50000, // Total fundraising milestone goal
    "raisedAmount": 31200, // Current verified donations tally
    "currency": "USD", // Currency: "USD" | "USDT" | "EUR" | "GBP" | "CAD"
    "showProgress": true, // Render progress bar in header & lobby
    "ctaLabel": "Fuel NextGen Athletes", // Call-to-action button label
    "ctaUrl": "https://nike.com/community" // Verified outbound link
  }
}
```

4. Tournament Operations & Administration REST API Catalogue

The backend must implement the following REST endpoints for tournament lifecycle operations:

| Method & Route                              | Auth Guard | Input DTO, Business Rules & Response Schema                                                                                                                                                                                                                |
|---------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET /admin/promo-tournaments                | AdminGuard | Returns complete roster of tournaments including entry counts, team breakdown, sponsor details, and active campaign state.<br><b>Response:</b> Array<SponsorTournament>                                                                                    |
| POST /admin/promo-tournaments               | AdminGuard | Creates a new tournament. If visibility=PRIVATE, generates collision-resistant 8-char sponsorCode & joinCode. If type=INFLUENCER, creates TournamentTeam rows with unique captainCode & memberCode.<br><b>Payload:</b> CreateTournamentDto                 |
| PATCH /admin/promo-tournaments/:id          | AdminGuard | Updates tournament parameters (title, description, GMT dates, time slot array, prizes, player caps).<br><b>Payload:</b> UpdateTournamentDto                                                                                                                |
| DELETE /admin/promo-tournaments/:id         | AdminGuard | Cascading deletion of tournament, teams, votes, player entries, campaigns, and asset records.                                                                                                                                                              |
| POST /admin/promo-tournaments/:id/status    | AdminGuard | Admin approves (status=APPROVED) or rejects (status=PENDING) a sponsor-submitted tournament.<br><b>Payload:</b> { action: 'approve'   'reject' }                                                                                                           |
| POST /admin/promo-tournaments/spons or-demo | AdminGuard | Generates instant, complete test arena (demo-moomorrow-cup), seeds fictional sponsor credentials, creates revision 1 campaign manifest, and outputs live URLs for instant partner demonstrations.<br><b>Response:</b> { tournament, sponsor, revision: 1 } |

|                                       |               |                                                                                                                                                                                                  |
|---------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET /promo-tournaments                | Public        | Public discovery list of active & upcoming tournaments (filters out private tournaments unless unlocked).<br><b>Response:</b> Array<PublicTournamentCardDto>                                     |
| GET /promo-tournaments/:id            | Public / Auth | Detailed tournament view. Accepts ?code=XYZ query param to unlock private tournament metadata.<br><b>Response:</b> TournamentDetailDto                                                           |
| POST /promo-tournaments/:id/join      | AuthGuard     | Registers player. Validates joinCode for private tournaments. If captainCode or memberCode is supplied, assigns player to corresponding TournamentTeam.<br><b>Payload:</b> { joinCode?: string } |
| POST /promo-tournaments/:id/vote-time | AuthGuard     | Casts or updates user vote for a GMT time slot. Recalculates vote aggregate and updates tally.<br><b>Payload:</b> { slot: '18:00' }                                                              |

## 5. Sponsor Campaign Studio, Media Upload Pipeline & Multi-Lane Approval System

The Campaign Studio implements rigorous enterprise release management for partner branding:

| Method & Route                                                    | Auth Guard     | Specifications, State Machine Transitions & Actions                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET /admin/promo-tournaments/:id/campaign                         | AdminOrSponsor | Returns complete campaign management payload: working draft manifest, active published revision, full version history with review comments, and asset catalog.                                                                                                                                                  |
| PATCH /admin/promo-tournaments/:id/campaign/draft                 | AdminOrSponsor | Saves changes into the working draft. If the latest version is locked (published or in-review), automatically bumps revision counter (e.g. 1 → 2) and creates a new DRAFT row.<br><b>Payload:</b> TournamentCampaignManifest                                                                                    |
| POST /admin/promo-tournaments/:id/campaign/assets/:kind           | AdminOrSponsor | Multipart file upload (LOGO, BACKGROUND_DESKTOP, BACKGROUND_MOBILE). Enforces MIME whitelist (image/png, image/jpeg, image/webp, image/svg+xml, video/mp4), 20MB limit, stores to disk or S3 bucket, creates TournamentCampaignAsset, marks previous active asset as archived, and returns updated asset array. |
| POST /admin/promo-tournaments/:id/campaign/submit-review          | AdminOrSponsor | Transitions draft status from DRAFT to IN_REVIEW. Locks draft from further editing until review decisions are recorded.                                                                                                                                                                                         |
| POST /admin/promo-tournaments/:id/campaign/versions/:vid/reviews  | AdminGuard     | Records independent approval decision for BRAND or SAFETY lane. When both lanes approve, version status automatically advances to APPROVED.<br><b>Payload:</b> { lane: 'BRAND' 'SAFETY', decision: 'APPROVED' 'CHANGES_REQUESTED' 'COMMENT', comment: string }                                                  |
| POST /admin/promo-tournaments/:id/campaign/publish                | AdminGuard     | Publishes the approved revision. Sets publishedAt = now(), marks previous published version as ARCHIVED, and purges Redis cache for immediate real-time propagation. Supports scheduling via activateAt / expireAt.                                                                                             |
| POST /admin/promo-tournaments/:id/campaign/versions/:vid/rollback | AdminGuard     | Instant 1-click rollback: clones specified historical version into a new published revision, bypassing draft cycle during emergency situations.                                                                                                                                                                 |
| POST /admin/promo-tournaments/:id/campaign/pause                  | AdminGuard     | Emergency circuit breaker: toggles isPaused. When true, public query returns { campaign: null }, immediately reverting the live match to the classic theme without dropping active players.                                                                                                                     |
| POST /admin/promo-tournaments/:id/campaign/cause/pause            | AdminGuard     | Freezes charity cause donations and progress indicators independently without disrupting visual sponsor branding.                                                                                                                                                                                               |

## 6. Campaign Templates System & 1-Click Deployment Engine

To support rapid partner demonstrations and reusable co-branding packages, the backend maintains the Template Registry in TournamentCampaignTemplate:

Standard Seed Presets in Database:

- 1. **Nike 31 · Speed & Youth Sports Fund:** Volt accent (#ccff00), Swoosh typography, \$50,000 USD goal for *Global Youth Athletic Alliance* (youth sports gear & community playgrounds).
- 2. **Adidas 31 · Ocean Plastic & Play:** Cyan accent (#38bdf8), Three-Stripes typography, €35,000 EUR goal for *Ocean Play Preservation Trust* (upcycled ocean plastic turf pitches).
- 3. **MooMorrow Pasture Charity:** Pasture green & gold (#8cff65), Farm-to-Table Fund, \$25,000 USD goal for local dairy farmers.
- 4. **Cyber Neon 31 · Code For Tomorrow:** Cyberpunk cyan/magenta (#67e8f9), NextGen Coder Initiative, 50,000 USDT goal for underprivileged school laptops.
- 5. **Golden Oasis Clean Water:** Gold/sky blue (#38bdf8), Clean Well Spring Foundation, 10,000 EUR goal for community wells.
- 6. **Obsidian Core Wildlife Rescue:** Emerald/flame (#f97316), Global Habitat Rescue, \$100,000 USD goal for endangered species.
- 7. **Candy Carnival Children's Smiles:** Hot pink/yellow (#facc15), Children's Smile Project, \$15,000 CAD goal for pediatric surgeries.

Endpoints to Implement:

- GET /admin/campaign-templates — Returns all 7 presets and custom templates.
- POST /admin/campaign-templates — Saves current studio customization as a named reusable template.
- DELETE /admin/campaign-templates/:id — Removes a custom template (system presets are protected).
- POST /admin/promo-tournaments/:id/campaign/apply-template — Atomic transaction: writes template manifest to draft, creates approved review record, and publishes version to the target tournament in a single step.

7. Dynamic [Sponsor] 31 In-Game Branding Engine & Real-Time Gateway Protocols

The platform dynamically resolves co-branded visual elements and syncs gameplay over WebSockets:

Public Query Resolution: GET /promo-tournaments/:id/campaign/active

- If isPaused=true, returns { campaign: null }.
- Resolves relative media paths to absolute CDN/API URLs.
- Frontend extracts manifest.logoTile.logoText || manifest.identity.sponsorName to render **[SPONSOR] [31]** in the header marquee (separate sponsor card + 3D golden '31' shield badge), the bottom arena HUD (ArcadeHUD), the mobile stats popup, and the lobby hero banner.

Real-Time Gateway Architecture (NestJS / Socket.IO / Redis Adapter):

- **Namespace:** /countdown with Redis Streams/Adapter for multi-instance horizontal scaling.
- **Events:**
  - room:join: Player joins room tour:{id}; server validates entry and returns current roster.
  - match:state\_sync: Broadcasts count (1–31), active turn player, timer (7s countdown), and last move played.
  - turn:move\_submit: Validates played cards (1–3 numbers), increments sum, resets turn timer, advances turn.
  - skill:activate: Validates skill cooldown, executes skill effect (Skip, Reverse, Freeze, Double), broadcasts visual trigger.
  - player:eliminated: Fired when player fails turn or exceeds count; triggers Barnaby mascot defeat animation.
  - match:game\_over: Fired when count reaches 31; crowns winner, awards prize pool USDT, updates leaderboard.

8. Privacy-Safe Telemetry, Analytics & Sponsor Proof-of-Performance Reporting

The platform aggregates privacy-safe metrics for sponsor ROI verification without tracking PII:

| Metric / Event                              | Trigger Condition & Ingestion                                                                            | Formula & Aggregation Logic                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Rendered Impressions<br>rendered_impression | Fired when sponsor header card, lobby hero, or cause badge enters viewport (50% pixels visible for ≥1s). | Count of verified impressions per placement and device class (desktop/tablet/mobile).                      |
| Viewable Seconds<br>viewable_seconds        | Batched 10-second dwell time heartbeat while match is active and browser tab is visible.                 | Total elapsed engagement seconds. Avg Dwell = Total Seconds / Impressions.                                 |
| Cause Expansions<br>cause_expand            | Fired when player clicks the Heart Cause button in the header to expand charity details.                 | Expansion Rate = Cause Expansions / Impressions * 100                                                      |
| CTA Click-Throughs<br>cta_click             | Fired when player clicks outbound link to sponsor or charity beneficiary website.                        | CTR = CTA Clicks / Total Eligible Impressions * 100                                                        |
| CSV Audit Streamer<br>GET /.../report/csv   | Sponsor or admin downloads proof-of-performance spreadsheet.                                             | Streams RFC 4180 CSV: Date, Revision, Placement, Device, Impressions, ViewableSec, Interactions, SharePct. |

## 9. Additional Production Requirements: Web3 Escrow, Shop & Background Jobs

### 1. Web3 Wallet & Solana USDT Escrow:

- **Deposit Processing:** Backend RPC listener monitors Solana SPL token transfers to the platform treasury wallet, matching deposits against DepositBinding.address to credit user USDT balances without race conditions.
- **Prize Pool Settlement:** Upon tournament completion, backend distributes prize pool USDT to top finishers via double-entry ledger (Wallet + LedgerEntry).

### 2. Background Crons & Scheduled Workers (BullMQ / NestJS Schedule):

- **Time-Vote Auto-Scheduler (Hourly):** Resolves winning GMT time slot 2 hours before startDate and locks startAt.
- **Campaign Telemetry Flusher (Every 60s):** Flushes in-memory/Redis telemetry batches into PostgreSQL aggregate table.
- **Scheduled Activations (Every 60s):** Checks for campaign versions with activateAt <= now() and publishes them automatically.

### 3. Security, Rate Limiting & Performance:

- **Distributed Rate Limiting:** Redis-backed throttling for login, turn moves, and telemetry ingestion.
- **Secret-Box Encryption:** TOTP 2FA secrets and sponsor recovery keys encrypted at rest using AES-256-GCM.
- **Cache Invalidation:** Redis key campaign:active:{tournamentId} invalidated on version publish/rollback.

## 10. Backend Developer Implementation & Verification Roadmap

### Phase 1: PostgreSQL Migrations & Prisma Schema

- [ ] Add TournamentCampaignTemplate model and verify relations in schema.prisma.
- [ ] Run `npx prisma migrate dev --name init_campaign_templates` and generate Prisma Client.
- [ ] Seed the 7 standard presets (Nike 31, Adidas 31, MooMorrow, Cyber Neon, Oasis, Obsidian, Candy).

### Phase 2: Controller & Service Implementation

- [ ] Implement SponsorsService & TournamentCampaignService with Zod/Class-Validator DTOs.
- [ ] Implement multipart file upload interceptor with disk storage / S3 stream in /admin/.../assets/:kind.
- [ ] Implement version bumping & review state machine (DRAFT → IN\_REVIEW → APPROVED → PUBLISHED).
- [ ] Implement 1-click template application endpoint (POST /admin/promo-tournaments/:id/campaign/apply-template).

### Phase 3: Real-Time Gateway & Telemetry Ingestion

- [ ] Wire CountdownGateway to emit live tournament start/vote notifications to connected sockets.
- [ ] Implement telemetry batch ingestion endpoint (POST /promo-tournaments/:id/campaign/events) with Redis queue buffering.
- [ ] Implement report CSV export streamer (GET /admin/promo-tournaments/:id/campaign/report/csv).

### Phase 4: Integration Testing & Verification

- [ ] Run end-to-end integration tests (`npm run test:e2e`) verifying multi-tenant security and zero-error frontend interaction.